

Question:

Four days after undergoing an elective total hip replacement, a 65-year-old woman develops a DVT that embolizes to the lung. Along with tachypnea, tachycardia, and cough, the patient would most likely present with a PaO2 of what?Answer Choices:A. 120 mmHgB. 110 mmHgC. 85 mmHgD. 60 mmHg

Reasoning + Answer:

<think>###FindingReasoningPaths: 1. **Surgicalstressandimmobility**âIncreasedvenousstasisâDVTâPulmonaryembolism (PE) âAcutehypoxemiaâLowPaO2. 2. **Pulmonaryembolism**âRightheartstrain (corpulmonale) âDecreasedoxygenationâHypoxemia. 3. **Tachypneaandtachycardia**âCompensatoryresponsetohypoxemiaâPaO2likelydecreased. 4. **Coughanddyspnea**âSymptomsofhypoxemiaâLowPaO2. 5. **CoagulationcascadeandcoagulationfactorXactivation**âClotformationâDVTâPEâAcutehypoxemia. ---###ReasoningProcess: 1. **Understandingthescenario**:Thepatientis a 65-year-oldwomanwhounderwentanelectivetotalhipreplacementfourdaysprior. Thisisaknownriskfactorforvenousthromboembolism (VTE), assurgery, particularlyofthelowerextremities, increasestheriskofdeepveinthrombosis (DVT).DVTisacommoncauseofpulmonaryembolism (PE),especiallyinthecontextofimmobilityduringtherecoveryperiod. 2. **PathophysiologyofDVTandPE**:DVTleadstotheformationofaclot (thrombus) inthedeepveinsofthelegs. Iftheclotbreakslooseandtravelstothelungs, itobstructspulmonarybloodflow, leadingtopulmonaryembolism. Thisdisruptsthenormalexchangeofoxygenandcarbondioxideinthelungs, resultinginhypoxemiaandassociatedsymptomssuchasdyspnea, tachypnea, andtachycardia. 3. **Clinicapresentation**:Thepatientpresentswithtachypnea, tachycardia, andaproductivecough. Thesesymptomsareconsistentwithhypoxemia, whichisadirectconsequenceofreducedoxygenationduetoPE. Therightheartstrain (corpulmonale) canoccuriftheembolismcauses significantobstruction, furtherimpairingoxygenation. 4. **Compensatoryresponse**:Tachypneaandtachycardiaarecompensatorymechanismstoincreaseoxygenationandperfusion. However, despitethese compensatoryresponses, thePaO2islikelytoremainlowduetotheunderlyingobstructioninthepulmonaryvasculature. 5. **ExpectedPaO2value**:Inthecontextofasignificantpulmonaryembolism, hypoxemiaisexpected. Typically, aPaO2below 80mmHg isconsideredabnormalandindicativeofsignificanthypoxemia. Theproductivecoughcouldbe secondarytoatelectasisorinfarctioninthelung, whichfurthercomplicatesoxygenexchange. ---###Conclusion: ThepatientwouldmostlikelypresentwithaPaO2 of**lessthan 80mmHg**, consistentwithseverehypoxemiadue totheacuteeffectsofpulmonaryembolism. </think><answer>85mmHg</answer>

Reward score